

## Parallel Port and Motor Operation

| Click button on screen | Pin 2 | Pin 3 | T1  | T2  | T3  | T4  | Motor rotation          |
|------------------------|-------|-------|-----|-----|-----|-----|-------------------------|
| Clockwise              | 1     | 0     | ON  | OFF | ON  | OFF | Clockwise               |
| Anticlockwise          | 0     | 1     | OFF | ON  | OFF | ON  | Anti-clockwise          |
| Brake                  | 1     | 1     | ON  | ON  | ON  | ON  | Motor experiences brake |
| Power OFF              | 0     | 0     | OFF | OFF | OFF | OFF | No power to motor       |
| Exit                   | —     | —     | —   | —   | —   | —   | Stop operation          |

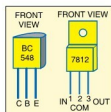


Fig. 6: Pin configurations of BC548 and 7812

shows the pin configurations of BC548 and 7812. The circuit comprises transistors BC548, opto-coupler MCT2E, regulator 7812 and a few discrete

components. H-bridge is formed using transistors T1 through T4, which are driven by the parallel port of a PC. Transistors act as switching elements as depicted in basic H-bridge configuration (Fig. 1). The opto-couplers IC1

and IC2 are used to provide physical isolation between the PC and the motor driver circuit. Here, pin 2 and pin 3 of the PC's parallel port are connected to pin 1 (anode of internal LED) of both the opto-couplers IC1 and IC2 through resistors R5 and R6, respectively. Resistors R5 and R6 are used to limit the current through internal LED of opto-couplers IC1 and IC2, respectively. Pin 18 through pin 25 of the PC's parallel port and pin 2 of opto-coupler IC1 and IC2 are grounded. Pin 5 (collector of internal transistor) of both the opto-couplers IC1 and IC2 are connected to 12V supply and pin 4 (emitter of internal transistor) controls the direction of the motor with the help of transistors. Resistors R1 through R4 are used to limit the base current of the transistors. The connector used is 25-pin D-type male connector. H-bridge and the parallel port should be grounded separately.

When pin 2 and pin 3 of the parallel port go high (logic 1) and low (logic 0) respectively, transistors T1 and T3 start conducting, while transistors T2 and T4 remain off. Current through the motor flows from left to right and the motor rotates in clockwise direction. When pin 2 and pin 3 of the parallel port go low and high, respectively, transistors T2 and T4 start conducting. The current flows through the motor in reverse direction and it rotates in anti-clockwise direction. When pin 2 and pin 3 of

the parallel port are set high, the motor stops immediately.

The braking facility is required in case of high-speed motors (or motors without gears), as they do not halt immediately due to inertia when the power supply is switched off. The table shows the parallel port, transistors and motor operation. Readers can use DC motor or DC geared motor as per requirement.

### PARTS LIST

#### Semiconductors:

- IC1, IC2 - MCT2E opto-coupler
- IC3 - 7812, 12V regulator
- T1-T4 - BC548 npn transistor
- D1-D5 - 1N4007 rectifier diode
- LED1 - 5mm LED

#### Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

- R1-R4 - 1.5-kilo-ohm
- R5, R6 - 330-ohm
- R7 - 680-ohm

#### Capacitors:

- C1 - 1000 $\mu$ F, 35V electrolytic
- C2 - 47 $\mu$ F, 25V electrolytic
- C3 - 0.1 $\mu$ F ceramic disk

#### Miscellaneous:

- X1 - 230V AC primary to 15V, 500 mA secondary transformer
- M1 - 12V DC motor
- 25 pin D-type male connector

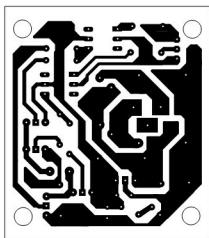


Fig. 7: Actual-size, single-side PCB for the H-bridge driver using PC

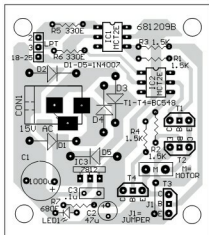


Fig. 8: Component layout for the PCB

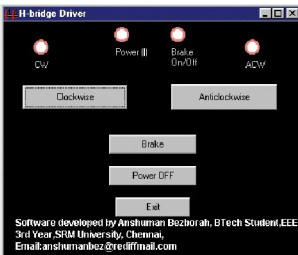


Fig. 9: Screenshot of main screen